



Faculty of Engineering Sciences and Technology
CURRICULUM FOR BS (SOFTWARE ENGINEERING) PROGRAM
(04 Years Degree Program)

Semester	Course(s) Code(New)	Course(s) Title	Credit Hours	Prerequisite
Semester I	SEN101	Applied Physics	3 + 1	---
	SEN111	Introduction to Information and Communication Technology	3 + 1	---
	SEN102	Calculus and Analytical Geometry	3 + 0	---
	HUM111	Functional English	3 + 0	---
	HMT121	Islamic Studies	2 + 0	---
	HMT122	Pakistan Studies	2 + 0	---
		Total Semester Credit Hours	(16 + 2)	
Semester II	BUS231	Financial Accounting Principles	3 + 0	---
	HUM231	Communication Skills	3 + 0	---
	SEN131	Programming Fundamentals	3 + 1	SEN111
	SEN103	Linear Algebra and Differential Equations	3 + 0	SEN102
	HUM112	Personal Development	3 + 0	---
		Total Semester Credit Hours	(15 + 1)	
Semester III	SEN201	Probability and Statistics	3 + 0	SEN102
	SEN231	Object Oriented Programming	3 + 1	SEN131
	SSC231	World History	3 + 0	---
	HUM233	Philosophy and Critical Thinking	3 + 0	---
	SEN241	Discrete Structures	3 + 0	---
		Total Semester Credit Hours	(15 + 1)	
Semester IV	HUM241	World Literature	3 + 0	---
	SEN232	Data Structures and Algorithms	3 + 1	SEN231
	HUM301	Sociology	3 + 0	---
	SEN221	Operating Systems	3 + 1	SEN231
	SEN251	Software Engineering	3 + 0	SEN231
		Total Semester Credit Hours	(15 + 2)	

Semester	Course(s) Code(New)	Course(s) Title	Credit Hours	Prerequisite
Semester V	SEN331	Database Management Systems	3 + 1	SEN231
	SEN351	Software Construction	3 + 0	SEN251
	SEN352	Human Computer Interaction	3 + 0	SEN251
		Elective I	3 + X	---
	HUM121	Academic & Professional Writing	3 + 0	---
	CSC333	Introduction to Python	0+1	SEN111
		Total Semester Credit Hours	(15 + 2+X)	
Semester VI		Elective II	3 + X	---
	SEN412	Information Security	3 + 0	SEN361
	SEN361	Data Communication and Computer Networks	3 + 1	SEN221
	SEN354	Software Requirement Engineering	3 + 0	SEN251
	SEN355	Web Engineering	3 + 1	SEN251
		Total Semester Credit Hours	(15 + 2+X)	
Semester VII	SEN451	Final Year Project – I	0 + 3	SEN251
		Elective III	3 + X	---
	SEN452	Software Re- Engineering	3 + 0	SEN251
	SEN453	Software Quality Engineering	3 + 0	SEN251
		Software Engineering – Domain Support I	3 + 0	---
		Elective IV	3 + X	---
		Total Semester Credit Hours	(15 + 3+X)	
Semester VIII	SEN454	Final Year Project – II	0 + 3	SEN451
	SEN455	Software Project Management	3 + 0	SEN251
	SEN456	Software Design and Architecture	3 + 0	SEN251
		Elective V	3 + X	---
		Software Engineering – Domain Support II	3 + 0	---
	HUM232	Ethics & Social Responsibility	3 + 0	---
		Total Semester Credit Hours	(15 + 3+X)	
Minimum Program Credit Hour(s): 137				

List of Elective Courses for BS (Software Engineering) Program

Course Code	Course(s) Title	Credit Hours
CSC336	Advanced Database Management Systems	3 + 0
CSC371	Digital Image Manipulation	3 + 0
CSC311	E-Business	3 + 0
CSC337	Web Programming Languages	3 + 1
CSC372	Digital Image Processing + Lab	3 + 1
CSC338	Visual Programming Languages + Lab	3 + 1
SEN358	Software Construction	3 + 0
SEN356	Software Requirement Engineering	3 + 0
CSC322	Robotics Programming + Lab	3 + 1
CSC362	Network Performance Management	3 + 0
CSC363	Network Security	3 + 0
CSC435	Mobile Application Development	3 + 0
SEN357	Software Evolution & Maintenance	3 + 0
CSC373	Computational Intelligence	3 + 0
CSC381	Computer Graphics & Animation + Lab	3 + 1
CSC436	Data Warehousing & Data Mining	3 + 0
CSC454	Enterprise Resource Planning	3 + 0
CSC474	Expert Systems	3 + 0
SEN455	Software Quality Assurance	3 + 0
SEN456	Enterprise Software Architecture	3 + 0
CSC475	Computer Vision	3 + 0
CSC463	Routing & Switching + Lab	3 + 1
CSC464	Wireless Communication + Lab	3 + 1
CSC465	Virtualization & Cloud Computing	3 + 0
CSC482	3D Modelling and Game Development + Lab	3 + 1
CSC466	Intranet Development	3 + 0

CSC312	Information System Audit	3 + 0
CSC476	Artificial Neural Networks	3 + 0
CSC477	Data Science	3 + 0
CSC478	Deep Learning	3 + 0
SEN457	Human Computer Interaction	3 + 0
CSC467	Internet of Things	3 + 0
CSC468	Software Defined Network + Lab	3 + 1
CSC469	Wireless Communication Networks	3 + 0
CSC437	Web Engineering + Lab	3 + 1
SEN458	Design Patterns	3 + 0
CSC423	FPGA Based System Design + Lab	3 + 1
CSC491	Bioinformatics	3 + 0
CSC424	Microprocessor Based Systems	3+1
CSC479	Machine Learning	3+0
CSC413	Technopreneurship	3+0
CSC445	Optimization Techniques	3+0
SEN359	Visual Software Analytics	3+0
SEN458	Management Information Systems	3+0
SEN459	Agent Based Software Engineering	3+0

List of Domain Support Courses for BS (Software Engineering) Program

Course Code	Course(s) Title	Credit Hours
SEN413	Business Process Engineering	3 + 0
SEN450	Formal Methods in Software Engineering	3 + 0
CSC446	Operations Research	3 + 0
CSC443	Simulation and Modeling	3 + 0
CSC444	Stochastic Processes	3 + 0

Course Area Category:

0-Maths & Science.

1-Information Technology.

2-Systems.

3-Programming & Databases.

4-Computer Science.

5-Software Engineering.

6-Networks & Communication.

7- Artificial Intelligence.

8-Graphics & Animation